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Name: ________________________________ May 6, 2017 Saturday

Department/School:_____________________ SID:____________________________________

University Physics I

Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1.(20pts.) A body of mass m is projected upwards with initial speed u and moves under uniform gravity and in a medium that exerts a resistance force that is proportional to the square of its speed. When the sphere falls vertically in the medium, its terminal velocity is v_0.

(a) Write the equation of the upward motion of the body.

(b) Find the maximum height the body attains and the time t_m taken to reach the maximum height.

(c) Write the equation of the downward motion of the body.

(d) Show that when the body returns to the starting point, its speed is $uv_0 \Big/ \sqrt{u^2+v_0^2}$.

(Hint: $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$.)

Solution: (a) Suppose the resistance force exerted by the medium is kv^2. When the body falls vertically with the terminal speed v_0, the acceleration is zero so that

$$mg - kv_0^2 = 0.$$

Hence, $k = \frac{mg}{v_0^2}$.

Therefore, for the upward motion, the equation of motion is

$$m\frac{dv}{dt} = -mg - kv^2 \quad \text{or} \quad \frac{dv}{dt} = -g\left(1+\frac{v^2}{v_0^2}\right).$$

(b)

Noting that $\frac{dv}{dt} = \frac{dv}{dz}\frac{dz}{dt} = v\frac{dv}{dz}$, the equation of motion can be rewritten as

$$v\frac{dv}{dz} = -g\left(1-\frac{v^2}{v_0^2}\right) \quad \text{or} \quad \frac{vdv}{v^2+v_0^2} = -\frac{g}{v_0^2}dz.$$

When the body reaches its maximum height, its velocity is zero. (It does not rise further more.) Then, integrating both side of the equation, we have

$$\int_u^0 \frac{vdv}{v^2+v_0^2} = -\frac{g}{v_0^2}\int_0^h dz \quad \text{or} \quad \frac{1}{2}\ln\frac{v_0^2}{u^2+v_0^2} = -\frac{g}{v_0^2}h.$$

Thus, $h = \frac{v_0^2}{2g}\ln\frac{u^2+v_0^2}{v_0^2}$ is the maximum height the body can attain.

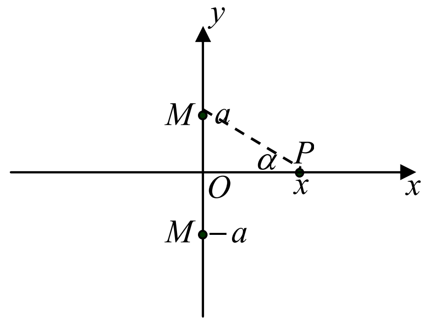
Rewriting the equation $m\frac{dv}{dt} = -mg\left(1+\frac{v^2}{v_0^2}\right)$ as

$$dt = -\frac{v_0^2}{g}\frac{dv}{v^2+v_0^2},$$

and integrating both sides, we have

$$t_m = \int_0^{t_m} dt = -\frac{v_0^2}{g}\int_u^0 \frac{dv}{v^2+v_0^2} = -\frac{v_0}{g}\left[\arctan\frac{v}{v_0}\right]_u^0 = \frac{v_0}{g}\arctan\frac{u}{v_0}.$$

(c) The equation of downward motion is



$$m\frac{dv}{dt}=mg-kv^2 \quad \text{or} \quad \frac{dv}{dt}=g\left(1-\frac{v^2}{v_0^2}\right).$$

(d) Rewriting the above equation of motion as $v\dfrac{dv}{dz}=g\left(1-\dfrac{v^2}{v_0^2}\right)$, separation of variables gives

$$\frac{vdv}{v_0^2-v^2}=\frac{g}{v_0^2}dz.$$

Integrating both sides of above equation, we obtain

$$-\frac{1}{2}\ln\left(v_0^2-v^2\right)=\frac{gz}{v_0^2}+C.$$

Using the initial condition $v=0$ for $z=0$, the integration constant C is

$$C=-\frac{1}{2}\ln v_0^2$$

Hence

$$z=-\frac{v_0^2}{2g}\ln\left(\frac{v_0^2-v^2}{v_0^2}\right).$$

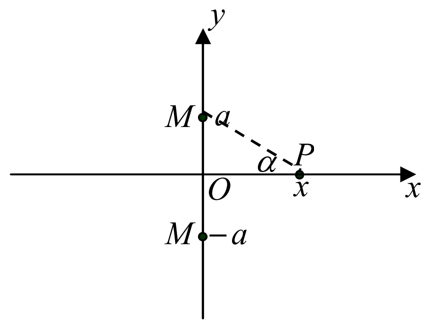
The body returns to its starting point when $z=h$, that is

$$-\frac{v_0^2}{2g}\ln\left(\frac{v_0^2-v^2}{v_0^2}\right)=\frac{v_0^2}{2g}\ln\frac{u^2+v_0^2}{v_0^2}.$$

Thus, we have

$$v=\frac{uv_0}{\sqrt{u^2+v_0^2}}.$$

2. (20pts.) A particle P of mass m is confined to move along the x-axis and is under the combined gravitational attraction of two particles, each of mass M, located at the points $(0,a,0)$ and $(0,-a,0)$ respectively.



(a) Show that the force field acting on P is given by

$$F(x)=-\frac{2GMmx}{\left(a^2+x^2\right)^{\frac{3}{2}}}.$$

(b) Find the corresponding potential energy function $V(x)$ of the force field of (a).

(c) Find the finite equilibrium point(s) of the particle P, and identify it/them as stable or unstable.

Solution: (a) By the law of gravitation, each particle of mass M exerts an attractive force on P and its magnitude is

$$F'=\frac{GMm}{R^2}$$

where $R^2=a^2+x^2$. By symmetry, the y-components of two attractive forces cancel each other and the resultant force points in the direction of $\overrightarrow{PO}$. Hence

$$F(x)=-\frac{2GMm}{R^2}\cos\alpha=-\frac{2GMm}{R^3}R\cos\alpha=-\frac{2GMmx}{\left(a^2+x^2\right)^{\frac{3}{2}}}.$$

(b) Taking the infinity as the reference point, the potential energy function is given by

$$V(x)=-\int_{\infty}^{x}F(x)dx=\int_{\infty}^{x}\frac{2GMmx}{\left(a^2+x^2\right)^{\frac{3}{2}}}dx=\int_{\infty}^{x}\frac{GMm}{\left(a^2+x^2\right)^{\frac{3}{2}}}d\left(x^2+a^2\right)=-\frac{2GMm}{\left(a^2+x^2\right)^{\frac{1}{2}}}.$$

(c) The condition for equilibrium is $F(x)=-\dfrac{2GMmx}{\left(a^2+x^2\right)^{\frac{3}{2}}}=0$. Thus $x=0$ is the equilibrium position.

Since $F(x)<0$ for $x>0$ and $F(x)>0$ for $x<0$, it is a restoring force around $x=0$. Therefore, the equilibrium is stable.

3. (20pts.) A uniform stick of length $2a$ has its lower end in contact with the smooth horizontal surface of a table. Initially the stick stands vertically and is at rest. With a tiny kick the stick topples over.

(a) What is the angular velocity of the stick when it hits the surface of the table?
(b) What is the center-of-mass velocity of the stick when it hits the surface of the table?

Solution: (a) Since the lower end of the stick can slide freely on the smooth horizontal surface of the table, there is no horizontal component of the forces acting on the stick when it falls down. Therefore, the horizontal component of the center-of-mass velocity is zero:

$$\dot{x}_C = 0.$$

Suppose θ is the angle between the stick and upward vertical. Then the vertical center-of-mass coordinate can be expressed as

$$z_C = a\cos\theta.$$

And the vertical component of the center-of-mass velocity is given by

$$\dot{z}_C = -a\sin\theta\dot{\theta}.$$

Therefore, the total kinetic energy of the stick can be written as

$$T = \frac{1}{2}m\left(-a\sin\theta\dot{\theta}\right)^2 + \frac{1}{2}I\dot{\theta}^2$$

where $I = \frac{1}{12}m(2a)^2 = \frac{1}{3}ma^2$. The gravitational potential energy of the stick is

$$V = mgz_c = mga\cos\theta.$$

The conservation of mechanic energy is

$$T + V = \frac{1}{2}ma^2\sin^2\theta\dot{\theta}^2 + \frac{1}{6}ma^2\dot{\theta}^2 + mga\cos\theta = C.$$

When $\theta = 0$, $\dot{\theta} = 0$, we have $C = mga$. When $\theta = \frac{\pi}{2}$,

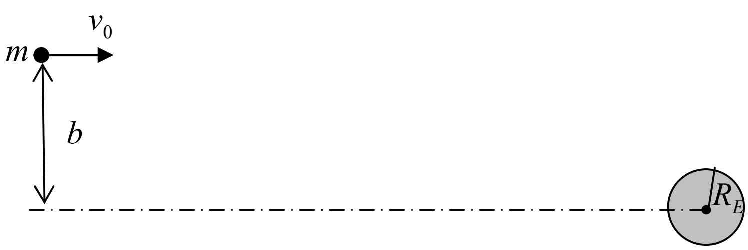
$$T + V = \frac{1}{2}ma^2\dot{\theta}^2 + \frac{1}{6}ma^2\dot{\theta}^2 = C = mga \quad \text{or} \quad \frac{2}{3}ma^2\dot{\theta}^2 = mga.$$

From above equation, we can easily obtain $\dot{\theta} = \sqrt{\frac{3}{2}\frac{g}{a}} = \frac{\sqrt{6}}{2}\sqrt{\frac{g}{a}}$. That is the angular velocity of the stick when it hits the surface of the table.

(b) The center-of-mass velocity of the stick when it hits the surface of the table is given by

$$\dot{z}_C = -a\sin\theta\dot{\theta} = -a\frac{\sqrt{6}}{2}\sqrt{\frac{g}{a}} = -\frac{\sqrt{6}}{2}\sqrt{ga}.$$

4. (20pts.) An asteroid of mass m approaches the Earth with impact parameter b and velocity v_0 that is relative to the Earth. The influences of the Sun and Moon on the asteroid can be ignored.



(a) What is the initial angular momentum of asteroid with respect to the center of the Earth?
(b) What is the closest approach r_A of the asteroid to the center of the Earth? Express the answer in terms of b, v_0, R_E and g where R_E is the radius of the Earth and g is the gravity on the surface of the Earth.
(c) If $b = 20R_E$, what is the condition for v_0 so that the asteroid will hit the Earth?

Solution: (a) $L = mv_0 b$.

(b) Assume that the velocity of the asteroid at the closest approach is v. Then, we have the conservation of angular momentum

$$mvr_A = mv_0 b \tag{1}$$

and the conservation of mechanical energy

$$\frac{1}{2}mv^2 - \frac{GM_E m}{r_A} = \frac{1}{2}mv_0^2 \tag{2}$$

Substituting $v = \frac{v_0 b}{r_A}$ into Eq.(2), we have

$$\frac{1}{2}m\frac{v_0^2b^2}{r_A}-\frac{GM_Em}{r_A}=\frac{1}{2}mv_0^2. \tag{3}$$

After some algebra, Eq.(3) can be rewritten as

$$\left(\frac{1}{r_A}\right)^2-\frac{2GM_E}{v_0^2b^2}\frac{1}{r_A}-\frac{1}{b^2}=0. \tag{4}$$

Thus

$$\frac{1}{r_A}=\frac{\frac{2GM_E}{v_0^2b^2}+\sqrt{\left(\frac{2GM_E}{v_0^2b^2}\right)^2+\frac{4}{b^2}}}{2}=\frac{GM_E}{v_0^2b^2}+\sqrt{\left(\frac{GM_E}{v_0^2b^2}\right)^2+\frac{1}{b^2}}. \tag{5}$$

(Another root is obviously inappropriate.) and

$$r_A=\frac{v_0^2b^2}{GM_E+\sqrt{(GM_E)^2+v_0^4b^2}}.$$

(6)

Taking $\frac{GM_E}{R_E^2}=g$, we have

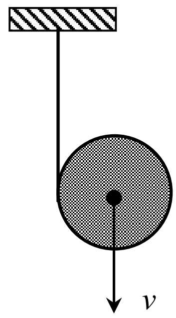
$$r_A=\frac{v_0^2b^2}{gR_E^2+\sqrt{\left(gR_E^2\right)^2+v_0^4b^2}}. \tag{7}$$

(c) When $r_A\le R_E$, the asteroid will hit the Earth. Substituting $b=20R_E$ into above equation, we have

$$r_A=\frac{400v_0^2R_E}{gR_E+\sqrt{(gR_E)^2+400v_0^4}}\le R_E$$

or $\quad v_0\le\sqrt{\frac{2gR_E}{399}}$.

5. (20 pts.) A uniform cylinder (a yo-yo) of radius R has a light inextensible string wrapped around it so that it does not slip. The free end of the string is secured to a fixed point and the yo-yo descends in a vertical straight line.



(a) Write the equation of translational motion of the center of mass and the equation of the rotational motion of the cylinder.
(b) Find the center-of-mass acceleration.
(c) If the cylinder is released from rest and it descends vertically down a height H, what are the center-of-mass velocity and the angular velocity of the cylinder?
(d) Deduce the results of (c) by adopting the conservation of mechanical energy approach.

Solution: (a) The equation of translational motion is

$$m\frac{dv}{dt}=mg-T. \tag{1}$$

The equation of rotational motion is

$$I\frac{d\omega}{dt}=TR,\qquad I=\frac{1}{2}mR^2. \tag{2}$$

The condition for rolling without slipping is $v=\omega R$ or $\omega=\frac{v}{R}$. (3)

(b) Substituting $I=\frac{1}{2}mR^2$ and $\omega=\frac{v}{R}$ into Eq. (3), we have

$$T = \frac{1}{2}m\frac{dv}{dt}. \tag{4}$$

Substituting Eq. (4) into Eq. (1), we have

$$m\frac{dv}{dt} = mg - \frac{1}{2}m\frac{dv}{dt}.$$

Therefore $\frac{3}{2}m\frac{dv}{dt} = mg$ and $a_c = \frac{dv}{dt} = \frac{2}{3}g.$

(c) $v = \sqrt{2a_c H} = \sqrt{\frac{4}{3}gH} = \frac{2}{3}\sqrt{3gH}$, $\omega = \frac{v}{R} = \frac{2}{3R}\sqrt{3gH}.$ (5)

(d) The conservation of mechanical energy gives

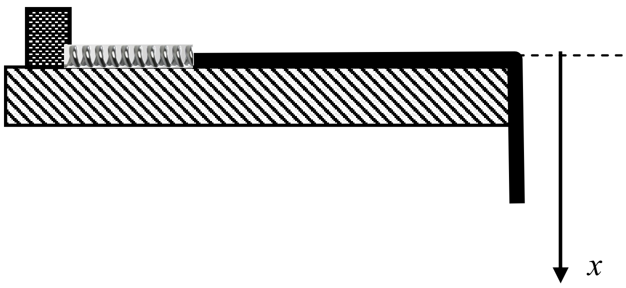
$$\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 - mgH = 0. \tag{6}$$

Substituting $I = \frac{1}{2}mR^2$ and $\omega = \frac{v}{R}$ into Eq. (6), we have

$$\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}mR^2\right)\left(\frac{v}{R}\right)^2 - mgH = 0 \quad \text{or} \quad \frac{3}{4}v^2 = gH.$$

Thus $v = \sqrt{\frac{4}{3}gH} = \frac{2}{3}\sqrt{3gh}$ and $\omega = \frac{v}{R} = \frac{2}{3R}\sqrt{3gH}.$

6. (20pts.) As shown in the figure, a light elastic spring of relaxed length l and spring constant k lies on a smooth horizontal table with one end fixed and the other attached to the left end of a heavy uniform flexible rope of mass M and length $4l$ .The other end of the rope hangs down over the edge of the table. When the spring has its relaxed length, the free end of the rope hangs a distance l vertically below the level of the table top. The system is released from rest in this position. The spring constant is $k = Mg/2l$,where g is the gravity.



(a)What is the equilibrium position of the free end of the rope?

(b)Show that the free end of the rope executes simple harmonic motion.

(c)Find the period of the simple harmonic motion of the free end of the rope.

Solution: (a) Supposing that the free end of the rope is distance x below the table top, the spring is stretched a displacement $x - l$. Then the condition for equilibrium is

$$k(x-l) = \rho xg$$

where $\rho = \frac{M}{4l}$ is the linear mass density of the rope. Substituting $k = \frac{Mg}{2l}$ and $\rho = \frac{M}{4l}$ into the equilibrium equation, we have

$$\frac{Mg}{2l}(x-l) = \frac{M}{4l}xg.$$

Thus $x_0 = 2l$ is the equilibrium position.

(b) The equation of motion is

$$M\ddot{x} = \frac{M}{4l}xg - \frac{Mg}{2l}(x-l) = -\frac{Mg}{4l}(x-2l).$$

Letting $\tilde{x} = x - 2l$, we have $\ddot{\tilde{x}} + \frac{g}{4l}\tilde{x} = 0$. Thus, the motion of the free end of the rope is a simple harmonic one around the equilibrium position $x_0 = 2l$.

(c) $\omega = \sqrt{\frac{g}{4l}}$, $T = \frac{2\pi}{\omega} = 4\pi\sqrt{\frac{l}{g}}.$